

Quiz 2 — Grading Key

M 375T Fall 2026 — TA copy. Q1: 2 pts; Q2: 4 pts; Q3: 4 pts. Partial credit: right idea / arithmetic slip loses 1 pt.

1. (i) $z = (8 - 6)/2 = 1$; (ii) $x = \hat{\mu} + \hat{\sigma}z = 6 + 2 \cdot (-1.5) = 3$. *Scoring: 1 pt each.* [2 pts]
2. (a) (50000,) — one entry per experiment; each entry counts how many of the 100 uniform samples in that row land in $[0, \frac{1}{2})$. (b) It estimates $\mathbb{P}(\text{exactly 50 of 100 samples land left}) = \binom{100}{50} 2^{-100}$ (≈ 0.0796 ; the expression suffices). *Scoring: 2 pts per part — in (a) one for the shape, one for the sentence; in (b) the exact expression is required for both points (the phrase alone earns 1).* [4 pts]
3. (a) $D(q||u) = \frac{1}{2} \log \frac{1/2}{1/4} + \frac{1}{2} \log \frac{1/2}{1/4} + 0 + 0 = \log 2$. (b) $e^{-nD} = e^{-100 \log 2} = 2^{-100}$ — astronomically small, so no: the uniform model cannot plausibly have produced this histogram; reject it. *Scoring: 2 pts for (a) ($\log 2$ or equivalent, e.g. $\ln 2 \approx 0.693$; 1 pt if the two zero terms are mishandled but the method is right); 2 pts for (b) — one for 2^{-100} , one for the verdict with a reason.* [4 pts]

Q1 tests standardization (Practice 3 A1); Q2 reading the uniform-coin idiom (Practice 3 A3/B2, Practice 2 A2 style); Q3 mirrors Practice 3 B4 (Sanov computation, via B3). Expect Q2(b)'s exact expression and Q3(a)'s zero bins to be the discriminators.